Summary in Graph

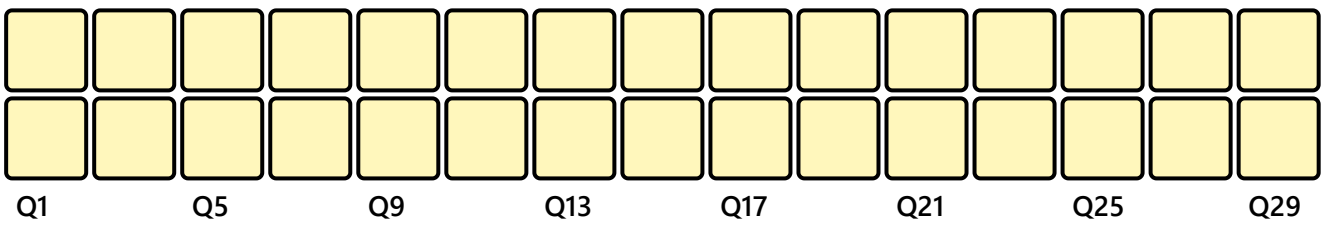
Exam Summary (GATE Overflow CS Test Series 2026 | Theory of Computation | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (7) out of 70 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Theory of Computation

Which of the following statements is/are TRUE regarding pumping lemma for regular languages? (Mark all the appropriate choices)

A. Satisfying pumping lemma is a necessary condition for a language to be regular
B. Satisfying pumping lemma is a sufficient condition for a language to be regular
C. Not satisfying pumping lemma is a necessary condition for a language to be not regular

D. Not satisfying pumping lemma is a sufficient condition for a language to be not regular

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Which of the following statements is/are TRUE? (Mark all the appropriate choices)

A. The set of all regular languages is countable
B. The set of all context-free languages is countable
C. The set of all recursive languages is countable
D. The set of all recursively enumerable languages is countable

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

If A is a regular language, E a deterministic context-free language, F a context-free language and for any language L, L^c denotes its complement, which of the following statements is/are TRUE? (Mark all the appropriate choices)

A. $E \cap F$ is context-free
B. $A \cap E^c$ is deterministic context-free
C. $E \cap F^c$ is context-free
D. $A \cap F^c$ is context-free

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Theory of Computation

The minimum number of states in the deterministic finite automata required to check the divisibility by 18 of a binary string (decimal value of string being divisible by 18) is ____

Your Answer: | Correct Answer: 10 | Not Attempted | Time taken: 00min 02sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Theory of Computation

For $L = \{\varepsilon\}$, if the number of total and final states required in the minimal state deterministic finite automata to accept L over $\Sigma = \{a, b\}$ are denoted by a and b respectively, then $a - b =$____

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

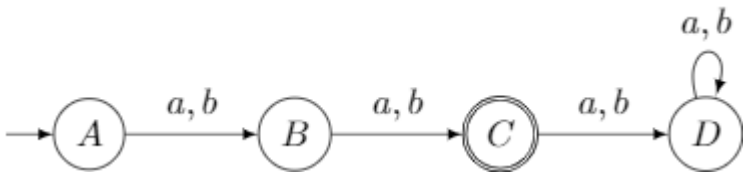
Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Theory of Computation

In the DFA (not necessarily minimal) for a regular language L over $\Sigma = \{a, b, c, d\}$, the minimum number of outgoing edges for any state is ____

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

The following deterministic finite automata recognizes

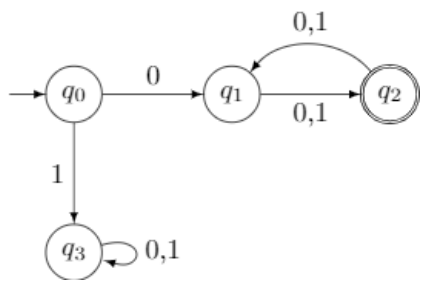


A. Set of all strings over $\{a, b\}$ of length exactly 2
B. Set of all strings over $\{a, b\}$ of length atleast 2
C. Set of all strings over $\{a, b\}$ of length atmost 2
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

The following deterministic finite automata recognizes



A. Set of all even length strings starting with 00
B. Set of all even length strings starting with 0
C. Set of all even length strings starting with 01
D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

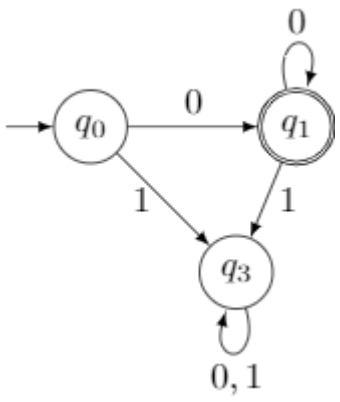
Q #9 Numerical Type | Award: 1 | Penalty: 0 | Theory of Computation

The number of two state DFAs with a designated initial state and a designated final state that can be constructed over the alphabet set $\Sigma = \{0, 1\}$ is _____

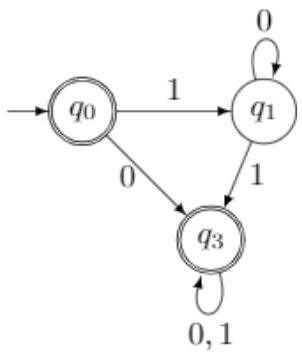
Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

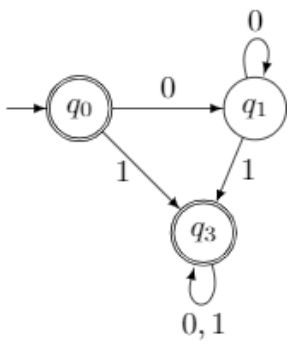
Which of the following is the complement of below Deterministic Finite Automata?



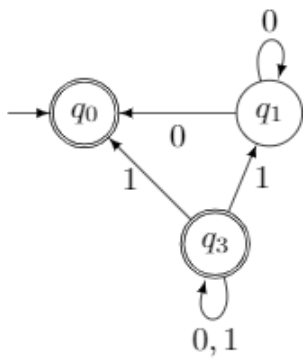
A.



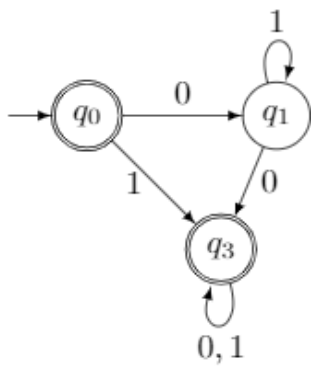
B.



C.



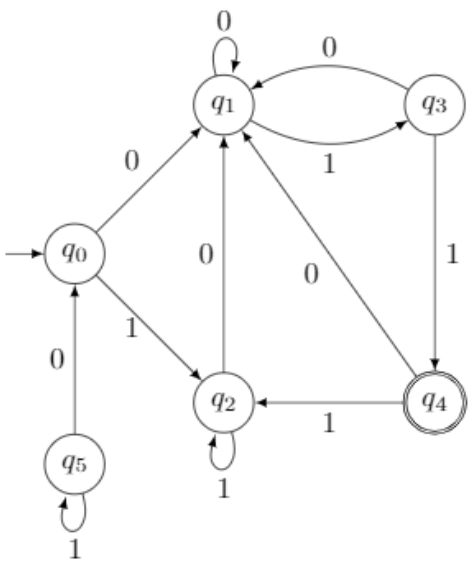
D.



Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Numerical Type | Award: 2 | Penalty: 0 | Theory of Computation

The number of states, after the minimization of the below Deterministic Finite Automata is _____



Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

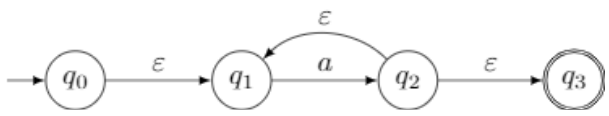
The regular expression for the language of all strings over $\Sigma = \{a, b\}$ with no consecutive $a's$ or $b's$ is given by ____ (Mark all the appropriate choices)

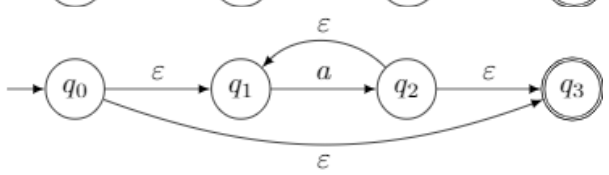
A. $(\varepsilon + b)(ab)^*(\varepsilon + a)$
B. $a(ba)^* + a(ba)^*b + b(ab)^*a + (ba)^*b$
C. $(\varepsilon + a)(ba)^*(\varepsilon + b)$
D. $a(ba)^* + a(ba)^*b + (ba)^* + (ba)^*b$

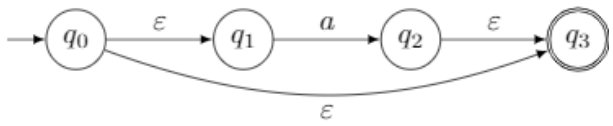
Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

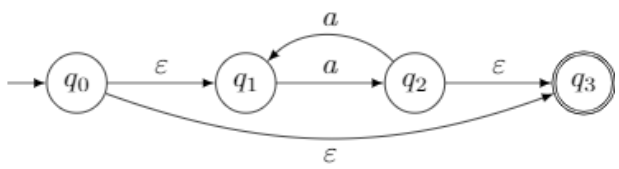
Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Given that $\Sigma = \{a\}$, which of the following finite automata correctly implements the regular expression a^*? (Mark all the correct options)

A.


B.


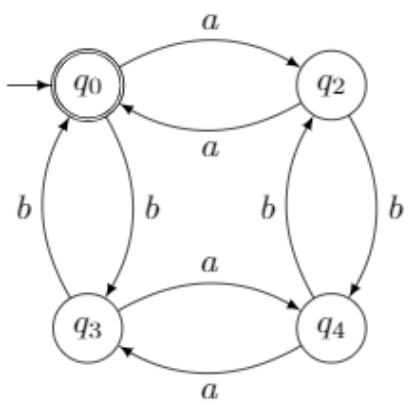
C.


D.


Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the given options correctly represents the finite automaton given below? (Mark all the appropriate choices)



A. $(aa + bb + (ba + ab)(ab + ba)^*(aa + bb))^*$
B. $(aa + bb + (ba + ab)(aa + bb)^*(ab + ba))^*$
C. $(ab + ba + (ba + ab)(aa + bb)^*(ab + ba))^*$
D. $(bb + aa + (ab + ba)(bb + aa)^*(ba + ab))^*$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

A language L has the prefix property if there are no two distinct strings in L, such that one is a prefix of the other. Which of the following language classes has/have prefix property? (Mark all the appropriate choices)

A. Finite Languages
B. Regular Languages
C. Deterministic Context Free Languages
D. Deterministic Context Free Languages accepted by Empty Stack Push Down Automata

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

A derivation of a string w of length n in a context-free grammar in Chomsky normal form

A. must involve exactly $2n-1$ applications of rules for all $n>0$
B. must involve at least $2n$ applications of rules for all $n>0$.
C. must involve at least n rules, but can involve an arbitrarily large number.
D. must involve at least n rules and at most $2n$ rules for all $n>0$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following is/are correct? (Mark all the appropriate choices)

A. For any CFG G there's a CFG G_0 such that G_0 is not ambiguous and $L(G)=L(G_0)$.
B. The CFLs are closed under symmetric difference
C. Every regular language can be generated by a CFG $G=(V,\Sigma,R,S)$ in which all rules are of the form $A\rightarrow aB$ or $A\rightarrow\varepsilon$ (where $A,B\in V$ and $a\in\Sigma$).
D. Regular languages having prefix property can be accepted by a Deterministic Pushdown Automata by empty stack

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Let L be the language containing only the single string w, where

$$w=\begin{cases}0 & \text{if India wins 2023 Cricket World Cup}\\ 1 & \text{otherwise}\end{cases}$$

Which of the following statements regarding L is TRUE?

A. L is regular
B. L is not recursive
C. L is not recursively enumerable
D. L is recursively enumerable but not recursive

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Mark all the context-free languages given below

A. $L = \{x\#w \mid x, w \in \Sigma^* \text{ and } x \text{ is a substring of } w\}$
B. $L = \{a^n b^{n+m} c^m d^n \mid n, m \geq 0\}$
C. $L = \{1^n \mid n \text{ is a power of } 2\}$
D. $L = \{0^n 1^m 0^m 1^n \mid m, n \geq 0\}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Which of the following strings is a member of the language over $\{a, b\}$ defined by the regular expression $(aa + ab)^*(bb)^*$?

A. $abbbba$
B. $abaabb$
C. $babbaa$
D. $bbaa$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the language L over $\Sigma = \{0, 1\}$ consisting of strings with equal number of $0's$ and $1s$. You are using the pumping lemma to show that L is not regular. You begin by supposing L were accepted by a DFA with k states. What would be a good choice of a string xyz that would allow you to obtain a contradiction?

A. 0011
B. 000111
C. $0^k 1^{2k}$
D. $0^k 1^k$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following are context-free grammar for the language that contains all strings (including empty string) with matched parenthesis?

A. $S \to (\,)$
$S \to SS$
$S \to (S)$
$S \to \varepsilon$
B. $S \to (\,)$
$S \to SS$
$S \to (S)$
C. $S \to (\,)$
$S \to SS$
$S \to \varepsilon$
D. $S \to (S)$
$S \to SS$
$S \to \varepsilon$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

What does the following context-free grammar describe?
S is the start symbol and the set of terminal symbols are $\{a, b\}$. The production rules are:

- $S \rightarrow a$
- $S \rightarrow aS$
- $S \rightarrow bS$

A. Strings that have an even number of $b's$ and $a's$
B. Strings that end in a
C. Strings that have an equal number of $b's$ and $a's$
D. Strings that have an odd number of $b's$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following is/are correct? (Mark all the appropriate choices)

A. A regular grammar is any right-linear or left-linear grammar.
B. $A \rightarrow \alpha A \mid \beta$ is a right linear grammar, where $\alpha, \beta \in T^*$, where T is the set of terminals
C. $A \rightarrow A\alpha \mid \beta$ is a left linear grammar, where $\alpha, \beta \in T^*$, where T is the set of terminals
D. Any grammar which generates a regular language, is a regular grammar

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the following language definition:
$L = \{\langle M \rangle \mid M \text{ is a DFA and } M \text{ accepts some string of the form } ww^R \text{ for some } w \in \Sigma^*\}$

L is

A. Regular
B. Context-free but not regular
C. Recursive but not context-free
D. Recursively enumerable but not recursive

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages is/are recursive? Here, $\langle M \rangle$ denote an encoding of Turing machine M. (Mark all the appropriate choices)

A. $L = \{\langle M \rangle \mid M \text{ moves its head left on input } w\}$
B. $L = \{\langle M, x \rangle \mid x \text{ is prefix of } \langle M \rangle\}$

C. $L = \langle M \rangle \mid M$ does not accept any string w such that 001 is a prefix of $w\}$
D. $L = \{\langle M \rangle \mid$ there exists a TM M_0 such that $\langle M \rangle \neq \langle M_0 \rangle$ and $L(M) = L(M_0)\}$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages is/are recursively enumerable but not recursive? Here, $\langle M \rangle$ denote an encoding of Turing machine M. (Mark all the appropriate choices)

A. $L = \{\langle M \rangle \mid L(M) = \emptyset\}$
B. $L = \{\langle M \rangle \mid L(M) = \Sigma^*\}$
C. $L = \{\langle M \rangle \mid |L(M)| = 2(\text{ number of strings in L(M) is } 2)\}$
D. $L = \{\langle M, w \rangle \mid w \in L(M)\}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages is/are not recursively enumerable? Here, $\langle M \rangle$ denote an encoding of Turing machine M. (Mark all the appropriate choices)

A. $L = \{\langle M \rangle \mid L(M) = \Sigma^*\}$
B. $L = \{\langle M \rangle \mid L(M) \text{ is finite}\}$
C. $L = \{\langle M \rangle \mid L(M) \text{ is infinite}\}$
D. $L = \{\langle M \rangle \mid L(M) = \emptyset\}$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages is/are regular? (Mark all the appropriate choices)

A. $L = \{xww_R \mid w, x \in (a+b)^+\}$
B. $L = \{xww \mid w, x \in (a+b)^*\}$
C. $L = \{xwyw \mid w, x, y \in (a+b)^+\}$
D. $L = \{wxw_R \mid w, x \in (a+b)^+\}$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

D_1 and D_2 are Deterministic Pushdown Automata and E is a Finite Automaton. Which of the following problems is/are decidable? (Mark all the appropriate choices)

A. Is $L(D_1) \cap L(D_2)$ a Deterministic-CFL?
B. Is $L(D_1) \cap L(D_2) = \emptyset$?
C. Is $L(D_1) \subseteq L(E)$?
D. Is $L(E) \subseteq L(D_1)$?

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

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